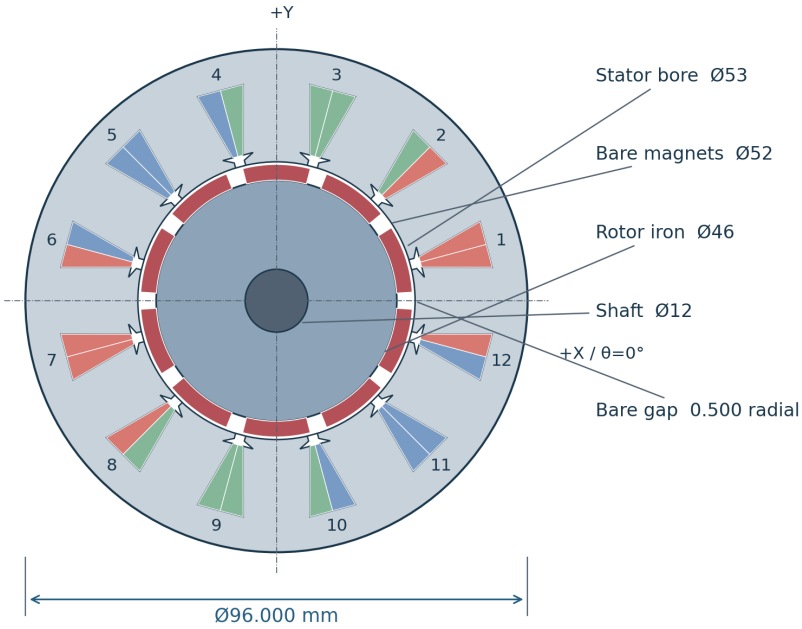
Motor dimensions and analysis

Three-phase permanent-magnet **inrunner** • 12 slots / 10 poles • Current 12-turn, two-strand winding

CAD starting point
Stator Ø96 × 85 stack, bore Ø53; rotor iron Ø46; bare magnet OD Ø52; basic shaft Ø12 × 130. The rotor and shaft turn inside the stationary stator.



Design quantity	Current value
Required motor shaft duty	4.365 N·m at 630 rpm
Associated actuator boundary	220 N·m at 10 rpm through assumed 63:1 / 80% gearbox
Electrical supply / current ceiling	42 V minimum, 48 V nominal / 22 A RMS phase-vector limit
Nominal motor prediction	10.834 A RMS; 15.734 V phase peak; 83.2% motor efficiency
Worst retained candidate thermal screen	119.1 °C buffered maximum versus 130 °C assumed limit

Assessment: the current motor is a plausible conditional match for the 220 N·m actuator duty. The optional 400 N·m case is not an accepted robust rating: later angle-resolved voltage and perturbation screens do not pass.

Pages 2–12 give dimensions, sketch points, winding tables and surrounding-part envelopes for CAD. Pages 13–19 consolidate the calculated performance and checks. Page 20 identifies the remaining CAD choices.

Exact active geometry is defined here. Finished bearing seats, endbells, coupling features, end-turn paths and retention construction remain design choices; no invented production dimensions are presented.

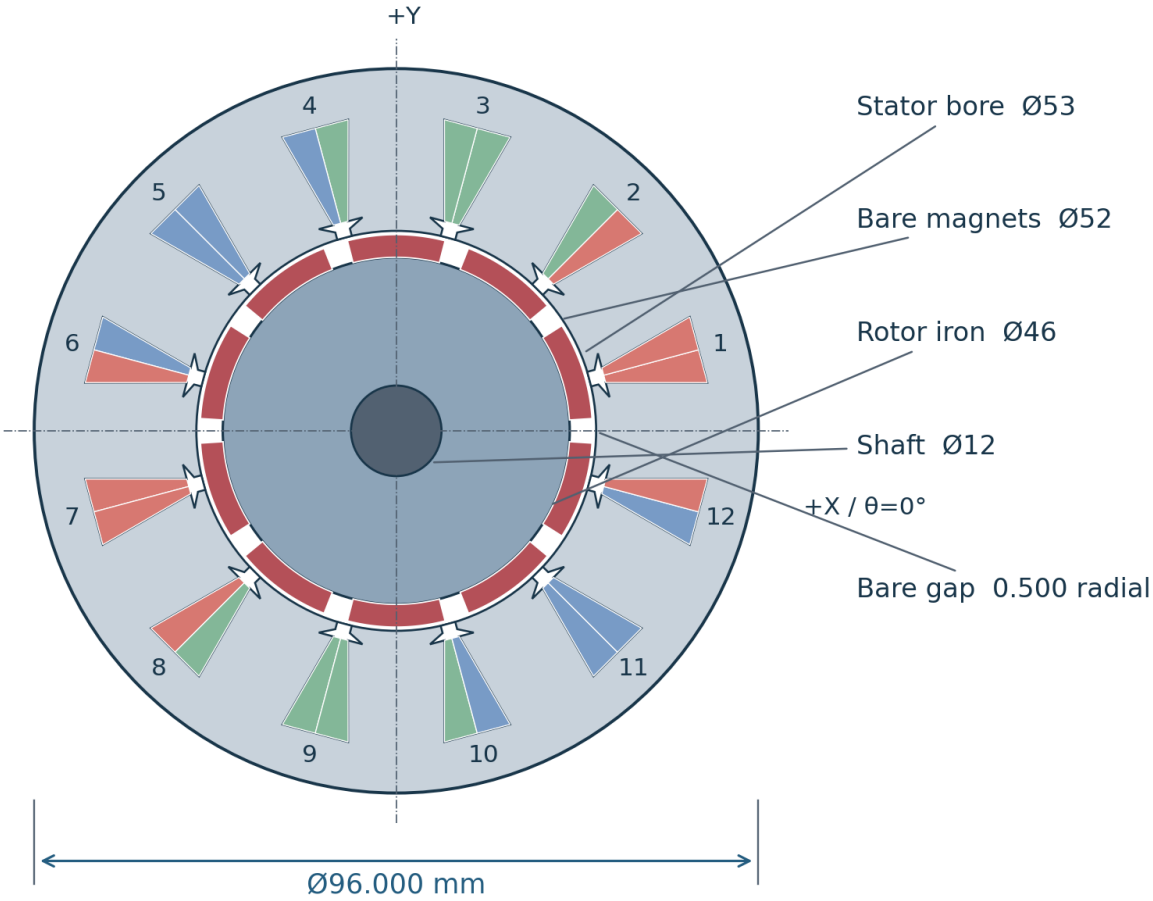
Master dimension schedule

MODEL = current native motor geometry; **DERIVED** = arithmetic from that geometry; **PROPOSED** = construction envelope, not part of the analyzed active CAD. All linear entries are mm.

Feature	Size / count	Basis
Stator outside diameter	96.000	MODEL
Stator bore diameter	53.000	MODEL
Stator / rotor / magnet active stack	85.000	MODEL
Rotor iron outside / inside diameter	46.000 / 12.000	MODEL
Bare magnet outside diameter	52.000	MODEL
Bare radial magnetic gap	0.500	DERIVED
Basic shaft diameter × length	12.000 × 130.000	MODEL envelope
Shaft overhang beyond active stack	22.500 per side, centered	Display datum only
Slots / poles / phases	12 / 10 / 3	MODEL
Slot pitch / pole pitch	30.000° / 36.000°	DERIVED
Magnet radial thickness / arc	3.000 / 28.800°	MODEL
Slot opening W0 / neck depth H0	2.000 / 1.000	MODEL
Slot shoulder W1 / height H1	6.000 / 1.000	MODEL; H1 is linear
Slot active depth H2 / back width W2	13.000 / 9.000	MODEL
Minimum stator radial back iron	6.275502	DERIVED; corner minimum
Lamination thickness / stacking factor	0.350 / 0.95	Material/model assumption
Turns per coil / series turns per phase	12 / 48	MODEL
Bare wire / radial enamel thickness	0.950 / 0.025	MODEL assumption
Strands in hand / coated wire OD	2 / 1.000	MODEL / DERIVED
Housing ID / OD / barrel length	96 / 112 / 85 nominal	PROPOSED; end features open
Housing wall	8.000	PROPOSED
Bearing bore × OD × width	12 × 32 × 10, two bearings	PROPOSED interface
Bearing-center span	113.000	Screen assumption
Radial magnet bond / retaining wrap	0.050 / 0.200	PROPOSED
Axial magnet segments per pole	4 × 21.250 before kerfs	PROPOSED; seams not allocated

Do not use the earlier 100 mm stator / 15-turn concept or the old 108 mm housing OD as current dimensions. Overall assembled motor length is not yet fixed by the basic 130 mm shaft.

Dimensioned active cross-section

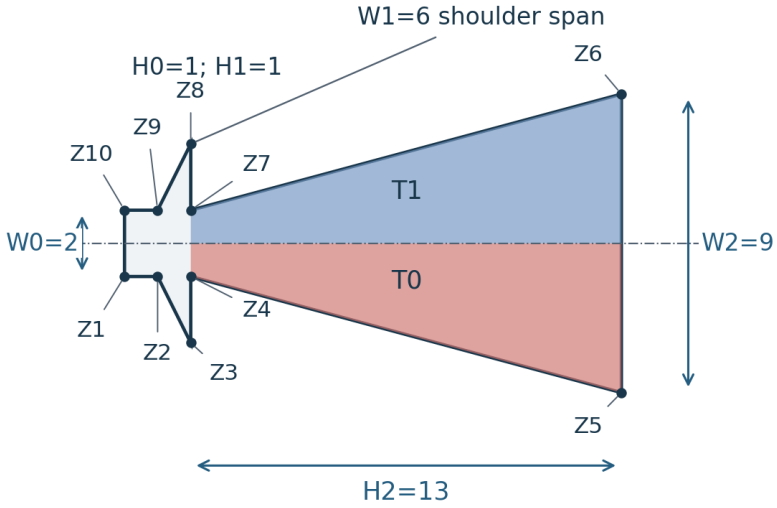


Reference	Definition
Origin and axis	Common rotor/stator center O=(0,0); shaft axis +Z
Viewing convention	XY coordinates as printed; +X at 0°, +Y at +90°; angles increase from +X toward +Y
Native slot 1 center	15°; other slot centers at 15° + 30°k, k=0…11
Native magnet 1 center	18°; other pole centers at 18° + 36°k, k=0…9
Axial extent	All active cross-section solids extruded from Z=0 to Z=85
Color / numbering	Steel gray, rotor blue-gray, magnets red, winding regions A/B/C colored; slot numbers 1…12

The native drawing has a 3° offset between the first slot center and first magnet center at its reference position. This is a geometric datum, not a calibrated encoder zero or phase-commutation instruction.

The 0.500 mm gap is from the bare Ø52 magnet envelope to the Ø53 bore. It does not already include the proposed adhesive build or retaining wrap. Those require the separate radial allocation on page 11.

Stator slot: exact sketch definition



Local coordinates below use the motor center as origin and a slot centered on +x. Connect Z1→Z2→…→Z10 with straight segments. Close along the short bore arc R26.5 from Z10 to Z1 through (26.5, 0).

Point	x, mm	y, mm
Z1	26.481125	-1.000000
Z2	27.481125	-1.000000
Z3	28.481125	-3.000000
Z4	28.481125	-1.000000
Z5	41.481125	-4.500000
Z6	41.481125	4.500000
Z7	28.481125	1.000000
Z8	28.481125	3.000000
Z9	27.481125	1.000000
Z10	26.481125	1.000000

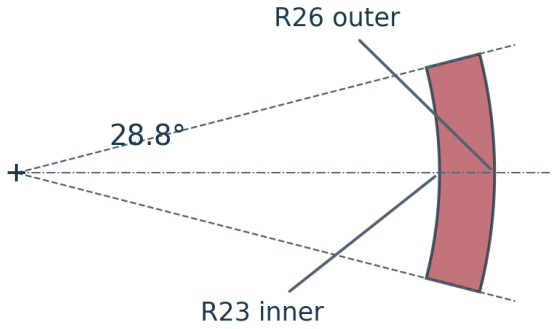
Parametric rule: $x(Z1)=\sqrt{(26.5^2-1^2)}=26.481125354$; $x(Z2)=x(Z1)+1$; $x(Z3)=x(Z4)=x(Z2)+1$; $x(Z5)=x(Z6)=x(Z4)+13$. The stepped shoulder defined by W1=6 is intentional; do not replace it with a simple trapezoid.

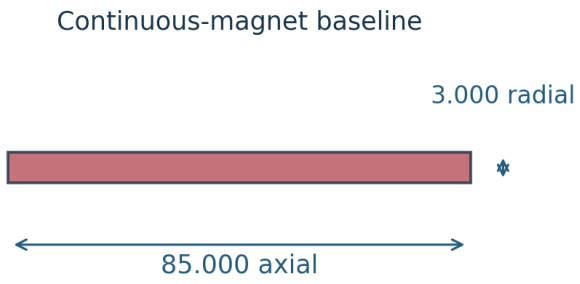
CAD operation: subtract the Ø53 bore and this slot void from the Ø96 disc. Rotate the first slot by +15°, pattern 12 copies at 30°, then extrude 85. Global points: $X=x\cos\theta-y\sin\theta$, $Y=x\sin\theta+y\cos\theta$.

Winding envelope: Z4→Z5→Z6→Z7 closes the active 71.500 mm² region. Divide it at local y=0 into T0 (y≤0) and T1 (y≥0), each 35.750 mm². Its depth is 13 mm. The outer slot-corner radius is 41.724498 mm, leaving 6.275502 mm minimum radial back iron.

Coordinate precision reconstructs the ideal model; the printed decimal places are not manufacturing tolerances.

Rotor and magnet dimensions





Four 21.25 mm segments are a separate proposal

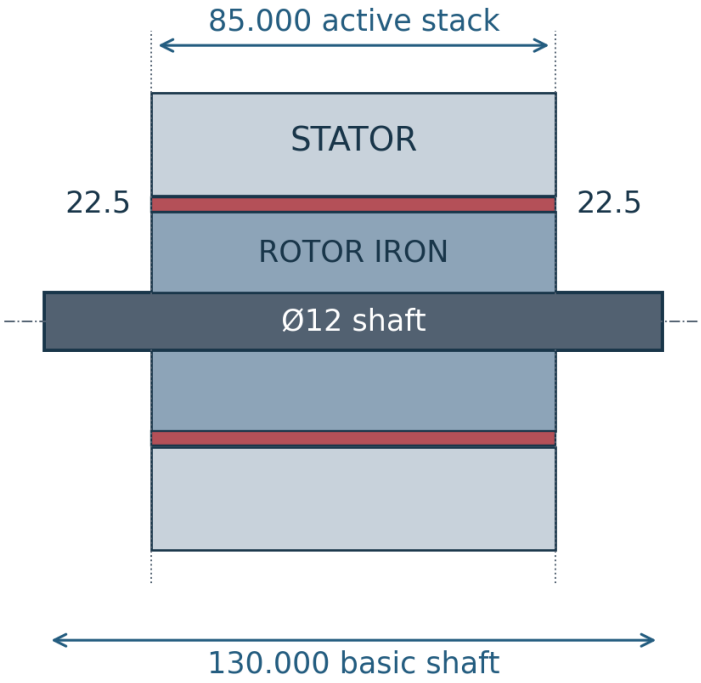
Part feature	Nominal dimension
Rotor steel	Annulus Ø46 / Ø12 × 85 axial
Each continuous magnet	Annular sector: inner R23, outer R26, 28.8° arc, 85 axial
Magnet thickness / pole pitch / inter-pole gap	3.000 radial / 36.000° / 7.200°
Outer / inner arc lengths	13.069025 / 11.561061
Outer / inner chord widths	12.931874 / 11.439735
Area / volume per continuous magnet	36.945130 mm² / 3140.336 mm³

Pole	Start °	Center °	End °	Pole	Start °	Center °	End °
1	3.6	18.0	32.4	6	183.6	198.0	212.4
2	39.6	54.0	68.4	7	219.6	234.0	248.4
3	75.6	90.0	104.4	8	255.6	270.0	284.4
4	111.6	126.0	140.4	9	291.6	306.0	320.4
5	147.6	162.0	176.4	10	327.6	342.0	356.4

Create one annular sector centered on 0° with edges ±14.4°, rotate it to 18°, then pattern 10 copies at 36°. Magnets alternate radial magnetization polarity. The absolute N/S reference must agree with the electrical calibration.

Segmentation proposal: four pieces per pole would nominally be 21.25 mm each only with zero axial seam/kerf. For three seams of width g, use piece length (85−3g)/4. The FE baseline uses continuous axial magnets; a segmented, bonded rotor is a construction revision.

Axial stack, shaft and datums



Longitudinal envelope view. Circumferential slot and magnet gaps are omitted. Dimensions refer to the unwrapped active model.

Feature	Z start	Z end	Length / note
Stator iron	0.000	85.000	85.000
Rotor iron	0.000	85.000	85.000
Continuous magnets	0.000	85.000	85.000
Active coil-side envelopes	0.000	85.000	85.000; end turns separate
Basic shaft	−22.500	107.500	130.000; Ø12 throughout model
Stack midplane	42.500	42.500	Reference for symmetric layouts

The basic shaft is a plain cylinder. Bearing-seat lengths, shoulder diameters, fillets, grooves, threads, keys/splines, rotor locking and the gearbox coupling are not defined by that cylinder.

The displayed 22.5 mm extensions are arithmetic from centering the 130 mm shaft on the 85 mm stack. They do not establish the final usable output extension, endbell positions or completed motor overall length.

Axial CAD constraint
Model the active stack at Z=0…85 first. Add real end-turn paths, bearings, endbells and coupling before freezing the finished shaft and overall assembly length. The 12 mm electrical end-length parameter is not a 12 mm axial overhang.

Winding and conductor dimensions

Winding quantity	Current definition
Topology	3-phase Y/star; double layer; tooth coil; coil pitch one slot
Coils / turns	12 coils total; 12 turns per coil; 4 series coils / 48 turns per phase
Parallel paths	1 phase circuit; two parallel strands in hand in every turn
Bare / insulated diameter per strand	0.950 / 1.000 mm; assumed enamel 0.025 mm radially
Bare copper area per strand / two-strand turn	0.708822 / 1.417644 mm²
One coil-side envelope / whole active slot	35.750 / 71.500 mm²
Conductors in one coil side / one full slot	24 / 48 individual wire cross-sections
Bare copper area per side / full slot	17.011724 / 34.023448 mm²
Bare fill / coated-wire area fraction	47.585% / 52.726% before liner, packing voids and insertion allowances
Phase resistance at 20 / 100 / 130 °C	0.127253 / 0.167261 / 0.182264 Ω
Line-to-line DC resistance at 20 °C	0.254506 Ω for Y connection, two phases in series

Electrical length allowance

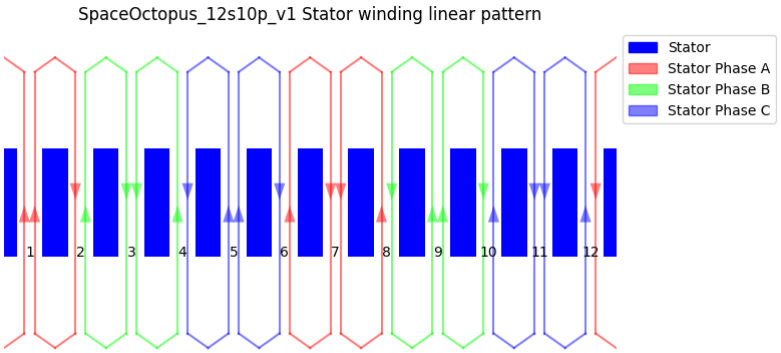
The native resistance model gives a modeled turn length of **218 mm**: 2×85 mm active length plus 4×12 mm end-length allowance. Each phase has 10.464 m of two-strand path; all six strand paths total 62.784 m, excluding external leads.

These lengths are resistance and mass allowances. They do not prescribe bend radius, end-turn axial projection, lead length or an individual wire route. Preserve the phase/polarity table when drawing the end turns.

CAD representation

For a first assembly, extrude the two half-slot quadrilaterals from page 4 through 85 mm. Assign the phase colors and polarities on page 8. Model individual Ø1.000 insulated strands only when the end-turn route and packing arrangement are chosen.

Slot map and coil pairing



Native winding pattern. Red=A, green=B, blue=C. Slot numbers here are human-readable 1…12.

Slot	Center °	T0 side	T1 side
1	15	A +12	A +12
2	45	A −12	B +12
3	75	B −12	B −12
4	105	B +12	C −12
5	135	C +12	C +12
6	165	C −12	A +12
7	195	A −12	A −12
8	225	A +12	B −12
9	255	B +12	B +12
10	285	B −12	C +12
11	315	C −12	C −12
12	345	C +12	A −12

T0 is the negative-local-y half-slot; T1 is the positive-local-y half-slot. These are tangential halves, not inner/outer radial layers. Native matrix indices are R=0 and T=0/1. “+12/−12” is signed turn count.

Phase	Four coils: positive side ↔ negative side
A	S1 T0 ↔ S12 T1; S1 T1 ↔ S2 T0; S6 T1 ↔ S7 T0; S8 T0 ↔ S7 T1
B	S2 T1 ↔ S3 T0; S4 T0 ↔ S3 T1; S9 T0 ↔ S8 T1; S9 T1 ↔ S10 T0
C	S5 T0 ↔ S4 T1; S5 T1 ↔ S6 T0; S10 T1 ↔ S11 T0; S12 T0 ↔ S11 T1

Each listed pair is one 12-turn coil. Connect all four coils of a phase in series with additive polarity, then join the three phase ends to the star point. Lead locations and physical series order are not specified. For CAD, use the shown signed axial directions as the reference convention.

Coil-side coordinates for CAD

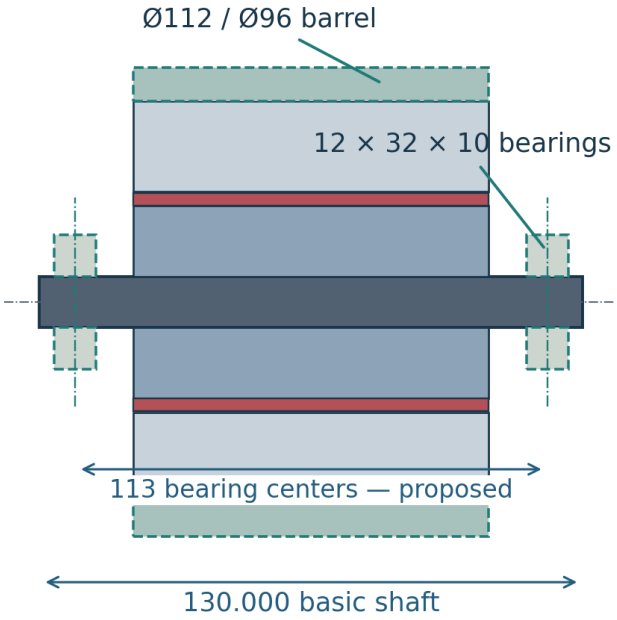
Each row identifies a point inside one active coil envelope. These are region reference points, not wire centers or boundary vertices. The slot sketch and y=0 division on page 4 define the boundaries.

Slot	Half	Phase	Turns	Reference X, mm	Reference Y, mm
1	T0	A	+12	34.145049	7.725633
1	T1	A	+12	33.433296	10.381929
2	T0	A	-12	25.707663	23.763119
2	T1	B	+12	23.763119	25.707663
3	T0	B	-12	10.381929	33.433296
3	T1	B	-12	7.725633	34.145049
4	T0	B	+12	-7.725633	34.145049
4	T1	C	-12	-10.381929	33.433296
5	T0	C	+12	-23.763119	25.707663
5	T1	C	+12	-25.707663	23.763119
6	T0	C	-12	-33.433296	10.381929
6	T1	A	+12	-34.145049	7.725633
7	T0	A	-12	-34.145049	-7.725633
7	T1	A	-12	-33.433296	-10.381929
8	T0	A	+12	-25.707663	-23.763119
8	T1	B	-12	-23.763119	-25.707663
9	T0	B	+12	-10.381929	-33.433296
9	T1	B	+12	-7.725633	-34.145049
10	T0	B	-12	7.725633	-34.145049
10	T1	C	+12	10.381929	-33.433296
11	T0	C	-12	23.763119	-25.707663
11	T1	C	-12	25.707663	-23.763119
12	T0	C	+12	33.433296	-10.381929
12	T1	A	-12	34.145049	-7.725633

Every active envelope extends from Z=0 to 85 mm. All 24 regions are separately identified; the model has 12 coils, not 24 coils. One full coil uses a positive and a negative active side joined by two end turns.

To reconstruct a boundary, rotate its local half-slot vertices by $\theta = 15° + 30°(\text{slot} - 1)$, then extrude along +Z. The exact boundary and this coordinate check make the winding assignment independent of a screenshot.

Housing and bearing envelope



Dashed green features are proposed envelopes. Bearing positions below are a derived symmetric CAD layout, not released seats or an analyzed 3D endbell.

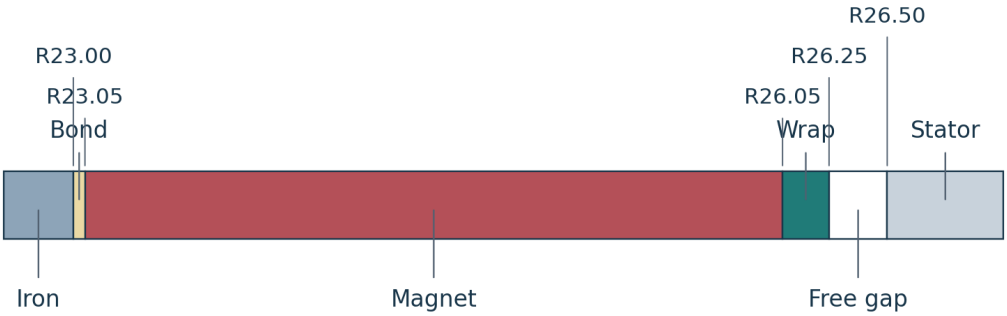
Proposed interface	Dimension / consequence
Aluminum housing barrel	Nominal ID96, OD112, wall 8, contact length 85
Housing-stator fit or bond clearance	Not selected; ID96 is a thermal-envelope reference, not a specified interference fit
Two bearing envelopes	Bore12 × OD32 × width10; catalog grade and preload not selected
Assumed bearing-center spacing	113
Symmetric center positions about Z42.5	Z=−14.000 and Z=99.000
Corresponding bearing axial envelopes	Z=−19…−9 and Z=94…104
Shaft extension beyond outer bearing faces	3.500 each with the present centered 130 mm cylinder
Distance stack face to nearest bearing face	9.000 per side in this proposed layout

The 3.5 mm remaining shaft end is not a selected coupling length. If end turns, retention, seals, encoder or coupling need more space, revise the shaft/endbell layout rather than treating 130 mm as a finished drawing.

Endbell thicknesses, fastening pattern/PCD, bolt sizes, seat tolerances, shoulders, preload hardware and lead exits remain open. Do not copy those sizes from the earlier 108 mm housing concept and assume they were checked here.

Cooling requires distributed contact along the barrel. An 8 mm wall with a mount only at one end does not reproduce the assessed boundary.

Magnet bond, wrap and radial gap



Proposed external-bond stack: all radial coordinates in mm

For an external 0.050 mm bond over R23 iron, keeping 3.000 mm magnet thickness gives proposed magnet radii 23.050…26.050 (OD52.100). A 0.200 mm wrap reaches R26.250 (OD52.500), leaving 0.250 mm cold radial clearance to R26.500 bore. This is a construction allocation, not the original bare-magnet CAD or a separately solved magnetic geometry.

Gap-budget term	Radial mm
Bare modeled radial gap	+0.500000
Worst screened thermal closure	−0.038220
Allocated radial magnet bond	−0.050000
Retaining wrap thickness	−0.200000
Total radial tolerance / initial-offset allocation	−0.050000
Wrap hoop expansion	−0.019994
Unsupported-span wrap bending	−0.013165
Additional shaft/bearing displacement	−0.016094
Remaining screened free gap	0.112527
Screen target / remaining margin	0.100000 / 0.012527

Proposed retention requirement	Value
Hot hoop modulus / strength with factor 2	100 GPa / ≥623.5 MPa
Hot interlaminar shear with factor 2	≥9.66 MPa
Compliant bond / hot shear modulus	0.050 mm / assumed 1 MPa
Required bond hot shear with factor 2	≥0.450 MPa; four axial segments

Only about 12.5 µm remains above the 0.1 mm gap target in this screen. The old 50 GPa wrap failed. Actual layup, fit, end retention and support stiffness must match the chosen CAD construction; do not silently double-count the 0.05 mm initial-offset allowance.

Materials and mass budget

Modeled component	Quantity / basis	Mass, kg
Stator lamination iron	12-slot stack; 0.95 stacking factor	2.534174
Winding copper	Electrical length includes modeled end allowance	0.398744
Rotor iron	Ø46 / Ø12 × 85; 0.95 factor	0.956755
Continuous magnets	10 sectors; assumed density 8300 kg/m³	0.260648
Stator + rotor modeled total	Excludes shaft and housing	4.150321
Basic shaft reference	Ø12 × 130; steel density 7650 kg/m³	0.112475
Proposed housing barrel	Ø112 / Ø96 × 85; assumed Al density	0.599868
Partial motor budget	Total above; bearings/endbells/etc. extra	4.862665

Material input	Value used / meaning
Electrical steel	Bundled M270-35A B–H exemplar; lamination 0.35 mm; density 7650 kg/m³
Steel mechanical assumptions	E=210 GPa; ν=0.30; stack factor 0.95
Steel loss reference	3 W/kg at 50 Hz, 1.5 T; assumed coefficients
Magnet recoil model	Generic SmCo-like; Br20=1.05 T; μr=1.05; αBr=−0.0003/K
Magnet conductivity basis	Resistivity 0.9 μΩ·m at 20 °C; assumed temperature coefficient
Copper electrical basis	$\rho 20 = 1.724 \times 10^{-8}$ Ω·m; αR=0.00393/K; density 8960 kg/m³
Housing thermal basis	Aluminum conductivity 167 W/(m·K); case capacity 600 J/K

The coil CAD solids are gross winding envelopes, not solid copper. Applying copper density to their entire volume would overestimate mass. The modeled winding mass includes its electrical end-length allowance, while the CAD envelopes include active length only.

The 4.863 kg partial budget is not a finished motor mass. Supplier magnet knee curves, high-field steel data and actual retention materials are not selected by naming a generic model.

Continuous motor operating results

The actuator boundary is 220 N·m at 10 rpm, ratio 63:1, efficiency 0.80. Thus required motor speed is 630 rpm, shaft torque is 220/(63×0.80)=4.365079 N·m, motor shaft power is 287.979 W, and actuator output power is 230.383 W.

Condition	Iq RMS A	Phase peak V	Copper W	Mean winding °C	Mean magnet °C
Running / sink50	10.834	15.734	51.79	59.64	58.57
Running / sink70 + loss stress	11.926	16.181	67.89	83.71	85.57
Holding / sink50	10.573	2.189	49.10	58.30	52.41
Holding / sink70	10.638	2.358	53.22	79.00	72.61

Nominal calculated item	Value
Electromagnetic torque / required shaft torque	4.464176 / 4.365079 N·m
Shaft mechanical power / actuator output power	287.979 / 230.383 W
Motor copper loss	51.789 W
Stator core / rotor core / magnet / proximity loss	3.3242 / 0.00662 / 0.2069 / 0.01659 W
Friction allowance	3.000 W at 630 rpm
Motor efficiency / system efficiency	83.15% / 64.53%
DC input / nominal 48 V average DC current	357.033 W / 7.438 A
Continuous copper current density	7.642 A/mm²

Electrical reference constants: at 80 °C magnets, RMS PM flux ≈27.279 mWb; local Ld≈0.905 mH and Lq≈0.902 mH, including assumed 80 µH end leakage. The ideal local PM torque term is ≈0.409 N·m per RMS ampere; electrical frequency at 630 rpm is 52.5 Hz. These local constants do not replace the nonlinear fault map.

Motor efficiency excludes the gearbox and inverter. The quoted DC input is the whole system budget at this duty, including the 80% gearbox boundary and assumed 97% inverter efficiency. The loss-stress scenario raises the non-copper loss allowance to 30 W and the sink to 70 °C; it is a declared sensitivity.

These winding temperatures are phase/volume averages. Local holding and pulse temperatures on page 15 are the relevant hot-spot screen; average temperature alone does not establish acceptance.

Voltage, torque margin and peak case

At 42 V, ideal linear SVPWM permits 42/√3 = 24.249 V phase peak. The design retains 10% reserve: **21.824 V phase peak**. Angle-resolved torque feedforward includes flux ripple and current-modulation voltage; it supersedes a mean-voltage-only peak screen.

Magnet °C	Output N·m	Drag W	Max Iq RMS A	Max phase peak V	Reserved budget
20	220	6.5	11.461	19.628	within
20	220	33.0	12.433	19.980	within
20	400	6.5	20.106	23.033	exceeded
20	400	33.0	21.079	23.440	exceeded
80	220	6.5	11.646	19.415	within
80	220	33.0	12.635	19.786	within
80	400	6.5	20.447	22.918	exceeded
80	400	33.0	21.438	23.335	exceeded
150	220	6.5	11.871	19.182	within
150	220	33.0	12.882	19.566	within
150	400	6.5	20.862	22.794	exceeded
150	400	33.0	21.874	23.223	exceeded

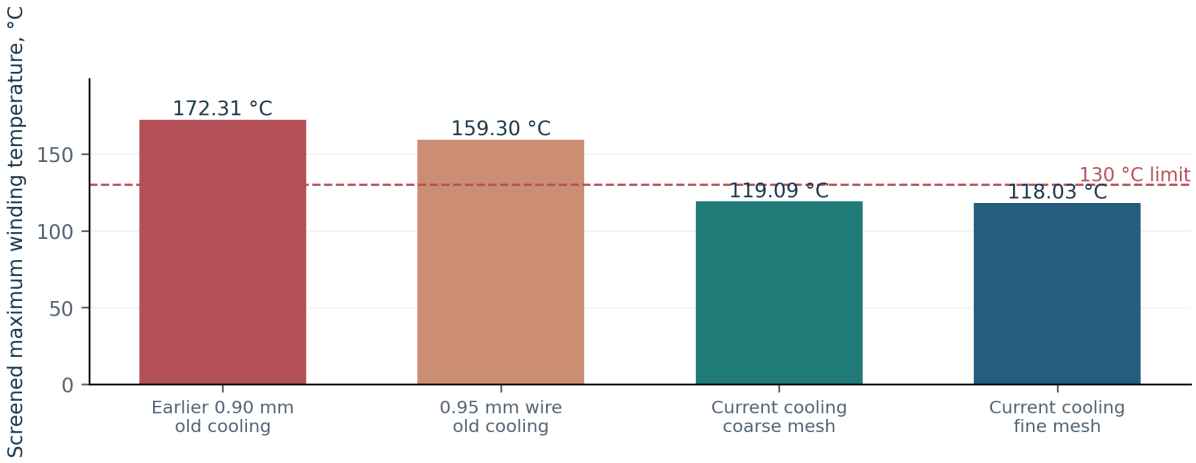
Hot 22 A RMS mean-torque perturbation screen

Perturbation at 150 °C magnets	EM torque N·m	Conditional output N·m
Br −5%	8.3681	396.5
Combined low flux	8.1845	387.3
Bare gap +0.050 mm	8.6637	411.4
Reference	8.8058	418.6
Stack −0.500 mm	8.7540	416.0

The output column subtracts 33 W motor drag, then applies the fixed gearbox ratio/efficiency. The combined case reaches only 387.3 N·m. Br −5% is below the specified 1.05 T minimum, not an accepted supplier tolerance. These are mean-torque scenarios, not a complete transmission rating.

CAD baseline to retain
Keep the 96 mm stator, 85 mm stack, 0.5 mm bare gap and 12-turn winding. Do not label the optional 400 N·m pulse as an accepted constant-output-torque rating.

Thermal results and cooling dimensions



Current candidate mesh	Steady max °C	Buffered 5 s max °C	Nodes
Coarse	104.82	119.09	11,696
Fine	103.75	118.03	24,941

Current results include spatial phase heating at sampled holding positions and conservative end-turn/case-energy offsets. The approximately 10.9 °C margin is conditional on the construction and losses below. It is not a measured temperature.

Required construction path / dimension	Adopted value
Assembled average winding-to-case resistance	≤0.120 K/W
Distributed full-barrel case-to-sink resistance	≤0.050 K/W
Housing / contact geometry	8 mm wall; nominal Ø112 outside; 85 mm contact length
Required outer-barrel contact conductance	≥692.7 W/(m²·K) in the assumed model
Proposed potting conductivity at 130 °C	≥0.400 W/(m·K)
Proposed slot liner thickness / conductivity	≤0.200 mm / ≥0.120 W/(m·K)
End-turn axial effective conductivity screen	25 W/(m·K), not a routed 3D coil solution
Winding / magnet temperature screen	130 °C / 150 °C
Radiation credit	None

The liner study used a zero-thickness thermal interface and did not remove its volume from the slot. The physical liner and packing must fit inside the page-4 envelope; the 47.6% bare fill is not a finished insulated-slot fit proof. A single end-flange mount does not satisfy the distributed cooling boundary.

Mechanical loads and support screen

Shaft torque N·m	Max shear MPa	Equivalent MPa	Twist ° over assumed span
4.500	13.263	22.972	0.2148
8.500	25.052	43.392	0.4058
13.416	39.541	68.488	0.6405

These are plain Ø12 solid-shaft torsion calculations with the 113 mm screen length. The 13.416 N·m row includes the studied fault load. Shoulder/fillet concentrations, fits, fatigue and actual coupling geometry are not represented.

Support / load item	Calculated result or assumed boundary
Rotor modeled mass	1.217403 kg, excluding the basic shaft
Simple supported first critical-speed estimate	22,629 rpm versus 630 rpm adopted duty
Static rotor offset in full-circle FE	0.050 mm; approximately 50 N mean magnetic pull
Magnetic negative radial stiffness	Approximately 1.0 MN/m
Assumed bearing-pair radial stiffness	10 MN/m in the retained gap screen
Combined shaft/bearing radial stiffness	4.156 MN/m in that support scenario
Additional offset amplification	Approximately 16.1 µm beyond the initial 50 µm
Bearing life scenario / required dynamic capacity	100 N load, 10,000 h; C ≥723 N

Native bore-pressure transfer to the stator

Support, fine mesh	Peak motion µm	Stress p99 MPa	First mode Hz
clamped	0.100	1.545	24447
spring_1e10	1.866	1.821	1459
spring_1e12	0.185	1.545	11011

The stator result is a 2D plane-stress/support idealization with 2% assumed damping. It is not a modal prediction of the complete housing, endbells, shaft and bearings. A 1 MN/m bearing-pair scenario failed the local magnetic-support stability screen; catalog preload/stiffness and real seats matter.

Fault current and magnet requirements

Winding / magnet °C	Motor rpm	Peak vector A RMS	Minimum Id A RMS
20 / 20	630	37.08	-33.10
20 / 20	945	43.16	-40.67
20 / 20	1890	51.46	-50.53
130 / 150	630	30.08	-24.77
130 / 150	945	36.36	-32.72
130 / 150	1890	45.74	-44.22

Varying assumed end leakage from 0 to 160 µH raises the largest sampled fault peak to **55.27 A RMS**. These currents exceed the 22 A RMS normal current limit. The nonlinear map-based fault model is not full transient electromagnetic FE.

Winding / magnet °C	rpm	Most adverse H parallel, kA/m	Required knee magnitude, kA/m
20 / 20	630	-565.0	678.0
20 / 20	1890	-710.3	852.3
130 / 150	630	-483.8	580.6
130 / 150	1890	-647.2	776.7

The supplier irreversible knee must remain more negative than the listed required magnitude at the corresponding temperature. The 20% screen allowance is included. **Irreversible-demagnetization acceptance is unset** because supplier curves are absent. These updated values supersede the earlier, smaller constant-inductance fault requirements.

Normal drive / material requirement	Value or status
Minimum / nominal bus; PWM frequency	42 / 48 V; 20 kHz studied controller
Normal vector RMS / instantaneous phase ceiling	22 A RMS / 31.113 A peak
Normal nonlinear PWM sampled peak vector current	≤21.557 A RMS in tested cases
Minimum assumed Br at 20 / 150 °C	1.050 / 1.00905 T
Maximum sampled fault magnetic traction	0.591525 MPa, used in retention screen

Actual current isolation, withstand and braking-energy hardware remain part of the drive design. Cold 22 A RMS native stator fields include local samples above the exemplar B–H curve's 1.8 T endpoint; use supplier high-field data before claiming peak material margin.

FEMM results and numerical verification

The working official electromagnetic route was Pyleecan 1.5.2 MagFEMM with FEMM 4.2. The main map has 27 nonlinear points: magnets 20/80/150 °C; Id=−12/−6/0 A RMS; Iq=0/10.814/22 A RMS. Independent intermediate holdouts were solved separately.

Full mechanical-period loss reference

Positions	Mean torque N·m	Stator core W	Rotor core W	Magnets W
144	4.42709	3.27593	0.006536	0.20510
288	4.42721	3.28440	0.006537	0.20443

Reference: 630 rpm, Iq=10.814 A RMS, Id=0, magnets 80 °C and loss resistance at 100 °C. The magnetic solve used the same slot/turn geometry; the reported copper resistance for the current candidate is on page 7. 144→288 positions changed magnet loss by 0.329%, rotor core loss by 0.018%. Separate spatial refinement changed non-copper loss components by less than 5%.

Hot loaded full-period check

Positions	Mean torque N·m	Stator core W	Rotor core W	Magnets W
144	8.80564	3.96520	0.021918	0.63176
288	8.80574	3.97951	0.021920	0.63118

Hot reference: 150 °C magnets, 130 °C winding, 22 A RMS, 630 rpm. The largest component loss change is 0.361%.

Independent check	Recorded outcome
Six normal-map holdouts	Max mean-torque error 0.0361%; flux-vector error 0.0761%
Twelve nonlinear fault-map holdouts	Evaluated independently from the fault-map grid
Isolated Ø0.95 wire FEMM AC benchmark	16 cases; fine-mesh exact-Bessel error ≤0.0322%
Incremental conductive-magnet AC response	20/40/80 kHz, two bias temperatures; 20 kHz mesh change ≤0.0145%
PWM incremental magnet-loss sensitivity	Approximately 0.083–0.112 W; not a validated 3D bound
Current CAD export check	37 valid solids; STEP import count preserved; volume error 3.30×10^{-13}

Numerical agreement is not measured hardware accuracy. Gmsh/scikit-fem/SciPy supplied identified project thermal, structural and dynamic couplings. FreeCAD was a project CAD export, not a claimed official Pyleecan coupling. Installed Elmer alternatives did not provide a completed solution for this motor.

Evaluated variants and baseline control

The following variants were evaluated; they are not silently included in the dimensioned baseline.

Actual nonlinear FEMM geometry grid

Magnet arc fraction	Bare gap mm	Mean torque N·m	Magnet mass kg	Ripple p-p N·m
0.7	0.4	4.2822	0.2281	0.0810
0.7	0.5	4.1472	0.2281	0.0799
0.7	0.6	4.0232	0.2281	0.0682
0.8	0.4	4.5684	0.2606	0.7316
0.8	0.5	4.4272	0.2606	0.6006
0.8	0.6	4.2938	0.2606	0.4836
0.9	0.4	4.7412	0.2932	0.4203
0.9	0.5	4.5933	0.2932	0.3353
0.9	0.6	4.4539	0.2932	0.2426

All rows use 10.814 A RMS and 80 °C magnets. The current baseline remains arc fraction 0.8 and bare gap 0.5 mm. Choosing another row changes geometry and does not inherit the complete baseline operating/fault/thermal assessment.

Optional two-slice rotor skew

Quantity	Unskewed reference	Two-slice option
Axial split / relative mechanical angle	None	2 × 42.5 mm / 3° relative
Mean torque at reference	4.42721 N·m	4.38924 N·m
Sampled torque ripple p-p	0.61538 N·m	0.06687 N·m
Ripple reduction	—	89.1% in the separate test

The 3° relative offset can be represented symmetrically as −1.5° / +1.5° in a new CAD variant. The complete unskewed operating map is not a full skewed map. Axial end effects, forces and construction of that stepped rotor remain outside the verified baseline.

The winding study compared 36 turn/wire choices and executed NSGA-II search. The adopted correction is 12 turns with two 0.95 mm strands and improved cooling. The 13-turn option and the earlier 0.90 mm wire are not the dimensions to use for this CAD baseline.

CAD build order and open dimensions

Step	CAD operation / drawing reference
1	Set mm units; common shaft axis +Z; active stack Z=0…85.
2	Create stator Ø96 / Ø53, cut the exact page-4 slot, rotate 15° and pattern 12×30°; extrude 85.
3	Create rotor annulus Ø46 / Ø12 × 85 and basic shaft Ø12 × 130 using page-6 display datums.
4	Create R23/R26 magnet sector, arc28.8°, center18°; pattern 10×36°; extrude 85.
5	Create the two half-slot winding envelopes; pattern 12 slots. Assign phases and polarity from pages 8–9.
6	Lay out the proposed Ø112/Ø96 housing and two 12×32×10 bearing envelopes. Check end-turn/coupling space before freezing endbells and shaft.
7	If adding bond/wrap/segmentation, revise geometry explicitly using page 11; do not overlap those solids with bare magnets.
8	Draw the chosen end turns and leads, then close the interfaces below and check assembled clearances.

Detail still to define in CAD	Why no final dimension is stated
Endbell thickness, fastening and bolt PCD	No completed motor enclosure drawing or mounting specification
Bearing fits, shoulders and preload hardware	Nominal bearing envelope only; no catalog/preload selection
Shaft seats, fillets and output coupling	Basic Ø12 cylinder is not a finished stepped shaft
End-turn route, projection and lead exits	Only topology and electrical end-length allowance are defined
Magnet segmentation seams and retention ends	Proposed bond/wrap screen does not set seam widths or end-lock geometry
Overall motor length and finished mass	Depend on the above surrounding parts and coil routing

What you can draw now

The stator, rotor iron, magnet sectors, active winding envelopes and basic shaft are dimensionally defined. Use the surrounding-part numbers as labeled proposals while completing the motor assembly.

This PDF consolidates the current executed motor study and later verification, with explicit model/proposal distinctions. It supersedes the need to browse the ZIP for basic dimensions. It does not claim absent assembly details were already designed.

Compilation: 6 November 2026. Data basis: current Pyleecan candidate, native winding/geometry exports, motor suitability report and additional nonlinear/FEMM verification. No new physical simulations were run to prepare this document.